SECR2043 OPERATING SYSTEMS

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Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	**Output**
	pwd	`naim@secr2043:~$ pwd` `/home/naim`
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	**Output**

	ls -a	`naim@secr2043:~$ ls -a`
		`.   .bash_history  .bashrc   .sudo_as_admin_successful  share`
		`..  .bash_logout   .profile  example.txt`

3.	Create a new directory named "os_lab" in your home directory.	
	Command	**Output**
	mkdir os_lab	No output
4.	Change your current working directory to "os_lab".	
	Command	**Output**

	cd os_lab	naim@secr2043:~$ cd os_lab naim@secr2043:~/os_lab$ pwd /home/naim/os_lab
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	**Output**
	touch hello.txt echo "hello, world" > hello.txt	no output
6.	Display the content of "hello.txt".	
	Command	**Output**
	cat hello.txt	naim@secr2043:~/os_lab$ cat hello.txt hello, world
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	**Output**
	cp hello.txt hello_copy.txt	no output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	**Output**
	mv hello_copy.txt hello_bak.txt	no output
9.	Delete "hello.txt".	
	Command	**Output**
	rm hello.txt	naim@secr2043:~/os_lab$ rm hello.txt naim@secr2043:~/os_lab$ ls hello_bak.txt
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	**Output**

	ls ~ -R -a -l -lh	naim@secr2043:~$ ls -R -a -l -lh .: total 32K drwxr-x--- 4 naim naim 4.0K Apr 22 07:11 . drwxr-xr-x 6 root root 4.0K Apr 22 06:20 .. -rw------- 1 naim naim 13 Apr 22 06:25 .bash_history -rw-r--r-- 1 naim naim 220 Apr 22 06:20 .bash_logout -rw-r--r-- 1 naim naim 3.7K Apr 22 06:20 .bashrc -rw-r--r-- 1 naim naim 807 Apr 22 06:20 .profile -rw-r--r-- 1 naim naim 0 Apr 22 06:25 .sudo_as_admin_successfu -rw-r--r-- 1 naim naim 0 Apr 22 06:57 example.txt drwxr-xr-x 2 naim naim 4.0K Apr 22 07:31 os_lab drwxr-xr-x 2 naim naim 4.0K Apr 22 06:57 share ./os_lab: total 12K drwxr-xr-x 2 naim naim 4.0K Apr 22 07:31 . drwxr-x--- 4 naim naim 4.0K Apr 22 07:11 .. -rw-r--r-- 1 naim naim 13 Apr 22 07:30 hello_bak.txt ./share: total 8.0K drwxr-xr-x 2 naim naim 4.0K Apr 22 06:57 . drwxr-x--- 4 naim naim 4.0K Apr 22 07:11 .. -rw-r--r-- 1 naim naim 0 Apr 22 06:57 textfile01.txt
	Total Mark:	

ACTIVITY 2 : **Text File Manipulation**

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

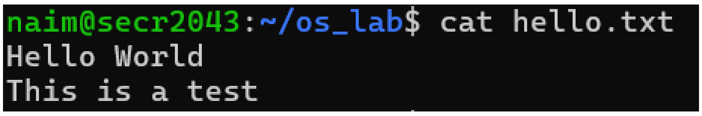
For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	**Output**
	touch hello.txt	naim@secr2043:~/os_lab$ touch hello.txt naim@secr2043:~/os_lab$ ls hello.txt hello_bak.txt
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	**Output**
	echo "Hello World" > hello.txt	no output

3. Append the text "This is a test" to "hello.txt" using echo.

Command	Output
echo "This is a test" > hello.txt	no output

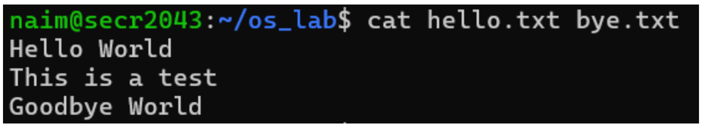
4. Display the contents of "hello.txt" using cat.

Command	Output
cat hello.txt	

5. Create another file named "bye.txt" with the text "Goodbye World" using echo.

Command	Output
touch bye.txt echo "Goodbye World" > bye.txt	

6. Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.

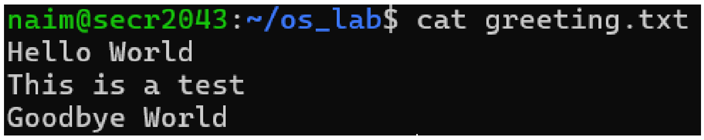
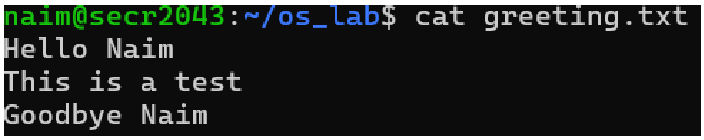
Command	Output
cat hello.txt bye.txt	

7. Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.

Command	Output
cat hello.txt bye.txt > greeting.txt	no output

8. Display the contents of "greeting.txt" using cat.

Command	Output
cat greeting.txt	

<table>
<tr><td></td><td></td><td>
naim@secr2043:~/os_lab$ cat greeting.txt
Hello World
This is a test
Goodbye World</td></tr>
<tr><td>9.</td><td colspan="2">Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>nano greeting.txt</td><td>
GNU nano 7.2 greeting.txt
Hello Naim
This is a test
Goodbye Naim
[Read 4 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify</td></tr>
<tr><td>10.</td><td colspan="2">Display the modified contents of "greeting.txt" using cat.</td></tr>
<tr><td></td><td>Command</td><td>Output</td></tr>
<tr><td></td><td>cat greeting.txt</td><td>
naim@secr2043:~/os_lab$ cat greeting.txt
Hello Naim
This is a test
Goodbye Naim</td></tr>
<tr><td></td><td>Total Mark:</td><td></td></tr>
</table>

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1. Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".

Command	Output

nano hello.c	GNU nano 7.2 hello.c *
	#include <stdio.h>
	
	int main() {
	printf("Hello World");
	return 0;
	}

2. Compile the program "hello.c" using gcc and generate an executable file named "hello".

Command	Output
gcc hello.c -o hello	no output

3. Run the executable file "hello" and display its output.

Command	Output
./hello	naim@secr2043:~/os_lab$ ./hello Hello World

4. Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".

Command	Output
nano sum.c	no output

5. Compile the program "sum.c" using gcc and generate an executable file named "sum".

Command	Output
gcc sum.c -o sum	no.output

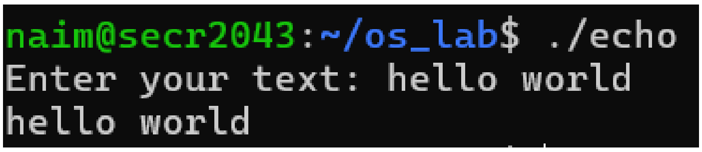
<table>
<tr><td rowspan="3">6.</td><td colspan="2">Run the executable file "sum" with 10 and 20 as arguments and display its output.</td></tr>
<tr><th>Command</th><th>Output</th></tr>
<tr><td>./sum 10 20</td><td><pre>naim@secr2043:~/os_lab$ ./sum 10 20
Sum: 30</pre></td></tr>
<tr><td rowspan="3">7.</td><td colspan="2">Run the executable file "sum" with -5 and 15 as arguments and display its output.</td></tr>
<tr><th>Command</th><th>Output</th></tr>
<tr><td>./sum -5 15</td><td><pre>naim@secr2043:~/os_lab$ ./sum -5 15
Sum: 10</pre></td></tr>
<tr><td rowspan="3">8.</td><td colspan="2">Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".</td></tr>
<tr><th>Command</th><th>Output</th></tr>
<tr><td>nano echo.c</td><td><pre>#include <stdio.h>

int main(int argc, char *argv[]) {
    if (argc < 2) {
        printf("Usage: %s \"<your text>\"\n", argv[0]);
        return 1;
    }

    char *text = argv[1];

    printf("%s\n", text);

    return 0;
}</pre></td></tr>
<tr><td rowspan="3">9.</td><td colspan="2">Compile the program "echo.c" using gcc and generate an executable file named "echo".</td></tr>
<tr><th>Command</th><th>Output</th></tr>
<tr><td>gcc echo.c -o echo</td><td>no output</td></tr>
<tr><td rowspan="3">10.</td><td colspan="2">Run the executable file "echo", type "Hello World" as input and display its output.</td></tr>
<tr><th>Command</th><th>Output</th></tr>
<tr><td>./echo</td><td></td></tr>
</table>

		 naim@secr2043:~/os_lab$ ./echo Enter your text: hello world hello world
	Total Mark:	